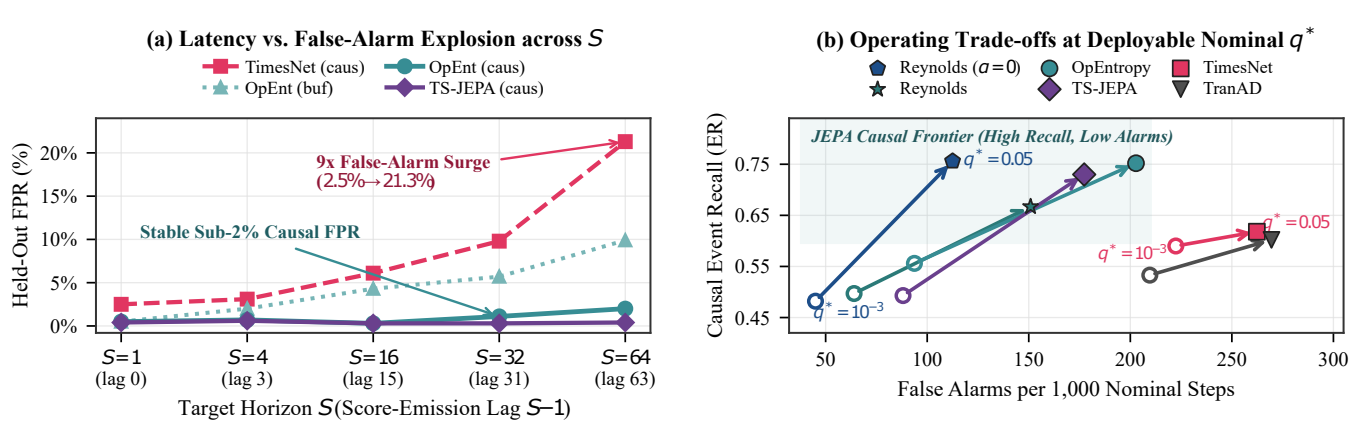

(a) Latency vs. False-Alarm Explosion across S
TimesNet (caus)
OpEnt (caus)
OpEnt (buf)
TS-JEPA (caus)
9x False-Alarm Surge
(2.5%→21.3%)
Stable Sub-2% Causal FPR
Held-Out FPR (%)
20%
15%
10%
5%
0%
S=1
(lag 0)
S=4
(lag 3)
S=16
(lag 15)
S=32
(lag 31)
S=64
(lag 63)
Target Horizon S(Score-Emission Lag S−1)
(b) Operating Trade-offs at Deployable Nominal q*
Reynolds (a=0)
Reynolds
OpEntropy
TS-JEPA
TimesNet
TranAD
JEPA Causal Frontier (High Recall, Low Alarms)
q* = 0.05
q* = 10−3
q* = 0.05
q* = 10−3
Causal Event Recall (ER)
0.75
0.65
0.55
0.45
50
100
150
200
250
300
False Alarms per 1,000 Nominal Steps